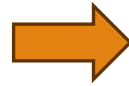


# Back-propagation algorithm

$$net_j = w_{j0} + \sum_i^n w_{ji} o_i, \quad o_j = f_j(net_j) \quad \delta_j = -\frac{\partial E}{\partial net_j} \text{ Credit assignment}$$

$$\Delta w_{ji} = -\alpha \frac{\partial E}{\partial w_{ji}} = -\alpha \frac{\partial E}{\partial net_j} \frac{\partial net_j}{\partial w_{ji}} = \alpha \delta_j o_i$$

$$\delta_j = -\frac{\partial E}{\partial o_j} \frac{\partial o_j}{\partial net_j} = -\frac{\partial E}{\partial o_j} f'(net_j)$$



If the jth node is an output unit

$$E = \frac{1}{2} (t_j - o_j)^2 \Rightarrow \frac{\partial E}{\partial o_j} = -(t_j - o_j)$$

$$\delta_j = (t_j - o_j) f'(net_j)$$

$$\frac{\partial E}{\partial o_j} = \sum_k^\kappa \frac{\partial E}{\partial net_k} \frac{\partial net_k}{\partial o_j} = -\sum_k^\kappa \delta_k w_{kj}$$

$$\delta_j = f'_j(net_j) \sum_k^\kappa \delta_k w_{kj}$$

$$\Delta w_{ji}(t) = \alpha \delta_j(t) o_i(t) + \gamma \Delta w_{ji}(t-1)$$

$$w_{ji}(t) = w_{ji}(t-1) + \Delta w_{ji}(t)$$

Momentum term to smooth  
The weight changes over time